

Exercises

Exercises on *foldr*

Exercise 1:

Using the higher-order function *foldr*, define a function *sumsq* which takes an integer n as its argument and returns the sum of the squares of the first n integers. That is to say, $\text{sumsq } n = 1^2 + 2^2 + 3^2 + \dots + n^2$.

Exercise 2:

Define *lengthr*, which returns the number of elements in a list, using *foldr*.

Exercise 3:

Define *myreverse*, which reverses a list, using *foldr*.

Exercise 4:

Using *foldr*, define a function *remove* which takes two strings as its arguments and removes every letter from the second list that occurs in the first list. For example,

```
remove "first" "second" = "econd".
```

Hint: Use a helper function in your lambda

Exercise 5:

The function *remdups* removes adjacent duplicates from a list. For example,

```
remdups [1, 2, 2, 3, 3, 3, 1, 1] = [1, 2, 3, 1]
```

Define *remdups* using *foldr* (and using a helper method).